

A Multivariate Permutation Test for Correlated Menstrual Cycle

Regularity Counts:

Application to Jewish Halachic Anticipatory Rules

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The dataset analyzed in this study was originally collected by Fehring et al. [6] and is publicly archived at Marquette University [7]. The author declares no conflicts of interest. Analysis code is publicly available at <https://github.com/dvirross/VesetVectorTest>.

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Abstract

Jewish religious law (*Halacha*) prescribes that women anticipate menstruation based on recurring patterns in their cycle history. These anticipatory rules are the *halachic vessel rules* (*vessel*: Hebrew for menstruation; the rules define the criteria by which a fixed menstrual pattern must be established and anticipated). Five of the 17 documented vessel rule types are testable with cycle-length data: *Haflaga* (fixed interval), *Dilug* (arithmetic progression), *Week* (weekly anchor), *Week-Dilug* (30-day anchor), and *Dilug-in-Dilug* (second-order progression). These form a polynomial hierarchy whose structure generates mathematically necessary dependencies, confirmed by exploratory pairwise permutation tests ($p_{\text{perm}} < .014$). We apply three complementary null models to 1,554 menstrual cycles from 118 women [6]: a global permutation null, a multinomial sampling null, and a within-woman permutation null conditioning on each woman's observed cycle-length multiset. Under the global null, *Haflaga* is strongly overrepresented ($z = 4.51$, $p < .001$; focused 3-pattern Mahalanobis joint test $D_M = 4.52$, $p < .001$), reflecting concentration of patterns among intrinsically regular women. Under the within-woman null — the scientifically sharper test — no pattern is significant (all $p \geq .722$; joint $D_M = 2.95$, $p = .121$), and all observed counts fall *below* their within-woman null means. These complementary results suggest that halachic vessel rules function primarily as empirical detectors of individual cycle regularity, with no detectable evidence, under within-woman exchangeability, that temporal ordering adds structure beyond each woman's own cycle-length distribution.

Keywords: menstrual cycle, vessel, Halacha, permutation test, within-woman permutation, multinomial sampling, natural family planning, Jewish law, polynomial hierarchy, dependence structure, multivariate randomization test

1. Introduction

Menstrual cycles exhibit substantial variability both within and across individuals [2, 10, 17]. The cycle length distribution is approximately bell-shaped with a pronounced right tail (skewness ≈ 1.3 in the present dataset), reflecting a small proportion of unusually long cycles. Within-person consistency is moderate: intraclass correlation coefficients for cycle length are typically in the range of 0.3–0.5 [3], indicating meaningful regularity beneath the surface variability. Bayesian hierarchical state-space models have been proposed to exploit this temporal structure for cycle-length prediction [1], confirming that within-woman dynamics carry genuine predictive information.

In the Jewish religious tradition, this regularity is also subject to detailed legal analysis. Halachic law (*Halacha*, Jewish religious law) requires that a woman anticipate the onset of her next menstruation based on patterns discernible in her prior cycle history. These anticipatory rules — the *halachic vessel rules* — govern a system of abstinence obligations central to the observance of *taharat hamishpacha* (family purity: the traditional Jewish system of laws governing marital intimacy in relation to the menstrual cycle). Of the 17 vessel pattern types documented in halachic literature, 10 require date or time-of-day data unavailable in standard cycle-length records; the remaining 7 are testable empirically. Two of these 7 (Dilug Chozer Chalila and Haflaga Chozer Chalila (Hebrew: repeating-cycle arithmetic progression and repeating fixed interval, respectively) occurred zero times in the present dataset and are not reported separately. The five patterns with positive observed frequencies are analysed here.

These five patterns form a *polynomial hierarchy*: each describes a *haflaga* (Hebrew: inter-menstrual interval, $H_k = L_k + 1$) sequence that follows a polynomial of a specific degree in cycle index. Haflaga, Week, and Week-Dilug are all degree-0 (constant) sequences; Dilug is degree-1 (linear); Dilug-in-Dilug is degree-2 (quadratic). This hierarchy generates mathematically necessary dependence relationships between the patterns, with implications for how their joint frequencies should be tested.

Despite the centuries-long practical application of vessel rules — and the modern availability of large longitudinal cycle datasets — no published study has formally tested whether these patterns occur in empirical data at rates exceeding chance. This is a non-trivial ques-

tion: if vessel rules merely detect random fluctuations, their anticipatory validity is weakened. Conversely, if they are significantly overrepresented, their empirical basis is independently corroborated.

The present study addresses this gap by applying two complementary null models — a permutation test [5] and a multinomial sampling null [16] — to a longitudinal NFP dataset [6]. We derive per-pattern p-values, effect sizes, and a joint multivariate test that accommodates the dependence structure among pattern counts without assuming independence. The joint test represents a methodological contribution [16], inspired by the simulation-based testing philosophy of King et al. [9] though distinct from their density-based approach; related ideas appear in multivariate permutation tests using quadratic forms [11]; a formal theoretical treatment is planned in future work. We also characterise the mathematical dependence structure through pairwise chi-square tests and derive the dependencies analytically from the pattern definitions.

This study extends a line of quantitative halachic scholarship [13–15]. To our knowledge it is the first to test vessel pattern frequencies against an explicit randomisation null, though we note this claim has not been verified by a systematic literature search.

2. Background

2.1 The Polynomial Hierarchy of Vessel Rules

Let $L_1, L_2, \dots, L_n$ denote the menstrual cycle lengths (in days) of a single woman, and define the *haflaga* $H_k = L_k + 1$ (the interval from the first day of menstruation k to the first day of menstruation $k + 1$, following halachic convention [14]). The five vessel pattern types are:

Haflaga. $H_k = H_{k-1} = H_{k-2}$ (three or more consecutive equal haflaga values, any value).

Degree-0 polynomial (constant, unrestricted). Minimum run: 3 cycles.

Dilug. $H_k - H_{k-1} = H_{k-1} - H_{k-2} = d \neq 0$ (constant non-zero first difference). Degree-1 polynomial (linear). Minimum run: 3 cycles.

Dilug-in-Dilug. $(H_k - H_{k-1}) - (H_{k-1} - H_{k-2}) = (H_{k-1} - H_{k-2}) - (H_{k-2} - H_{k-3}) = d \neq 0$ (constant non-zero second difference). Degree-2 polynomial (quadratic). Minimum run: 4 cycles.

Week. $H_k = H_{k-1}$, $H_k \in \mathcal{W}$ where $\mathcal{W} = \{H_{\min} + 3, H_{\min} + 10, \dots\}$ (values spaced 7 apart, corresponding to the same weekday at each menstruation). Degree-0 polynomial anchored to weekly values. Minimum run: 2 cycles.

Week-Dilug. $H_k = H_{k-1} = 30$ (two or more consecutive haflaga values both equal to 30). The value $H = 30$ (29 actual days = 4 weeks +1 day) is the unique haflaga for which each successive menstruation falls one weekday later, producing an arithmetic progression of onset weekdays — hence the name *Week-Dilug* (Hebrew: *shavu'a be-dilug*, week with arithmetic progression; [16]). Week-Dilug thus combines the constant-interval property of Haflaga with the arithmetic-progression concept of Dilug, applied to the weekday domain rather than the cycle-length domain. Degree-0 polynomial (constant) at the canonical 30-day anchor. Minimum run: 2 cycles.

The hierarchy Haflaga (degree 0) $\rightarrow$ Dilug (degree 1) $\rightarrow$ Dilug-in-Dilug (degree 2) mirrors the classical Taylor expansion: a constant function is the zeroth-order approximation, a linear function the first-order, and a quadratic the second-order. Week and Week-Dilug are specialisations of the degree-0 (constant) case anchored to calendar values.

2.2 Mathematical Dependence Structure

The polynomial hierarchy generates two analytically derivable dependence relationships between the remaining pattern pairs, which we verify empirically in Section 4.6:

Haflaga and Dilug-in-Dilug — proximity in the hierarchy. — When the second difference $|d|$ of a Dilug-in-Dilug sequence is small, its first differences are nearly equal and the haflaga sequence is nearly constant — i.e. nearly a Haflaga. In the data, 16 of 27 observed DiD sequences have $|d| = 1$, making them almost indistinguishable from a constant sequence. Both patterns further share the same structural prerequisite: a minimum run of ≥ 3 consecutive cycles satisfying a regularity condition, selecting the same women with high within-person cycle stability (within-person SD: Haflaga women = 1.54 days vs. no-vessel women = 3.17 days).

Dilug and Week-Dilug — logical incompatibility. — Dilug requires $\Delta H_k \neq 0$ (strictly non-constant differences); Week-Dilug requires $H_k = H_{k-1} = 30$ ($\Delta H = 0$). These conditions are mutually exclusive within the same cycle window, creating a negative dependence tendency.

2.3 Prior Work

Menstrual cycle statistics. — Variability and regularity of cycle length have been extensively characterised [2, 3, 10, 17]. Bortot et al. [1] developed a Bayesian hierarchical state-space model for sequential prediction of cycle lengths, exploiting within-woman temporal dependence; their model provides a benchmark for cycle-length predictability.

Quantitative halachic analysis. — Ross [15] modelled cycle transitions with Markov chains; [13] evaluated reliability of self-reported menstrual data in a halachic context; [14] applied statistical analysis to abstinence laws. The present study adds the first formal hypothesis test of whether vessel patterns are overrepresented in real data.

Randomization-based testing. — Permutation tests [5, 8] provide exact finite-sample p-values under exchangeability. The multinomial sampling null [16] draws each woman’s cycles independently from the fitted empirical distribution, directly encoding the null hypothesis of cycle independence. For the joint test, King et al. [9] proposed a general framework for testing a vector of statistics via simulation of the joint density, using a multivariate kernel density estimator to define a smallest-acceptance-region test. The present joint test differs: we use the Euclidean distance from the null mean vector as the test statistic, which is simpler, requires no density estimation, and has an intuitive geometric interpretation. A formal theoretical comparison and analysis of the distance-based approach is planned in future work [16].

3. Methods

3.1 Dataset

We analyzed a publicly available longitudinal dataset originally collected by Fehring et al. [6] in a randomised trial comparing two internet-supported natural family planning methods, archived at Marquette University [7]. After excluding cycles shorter than 18 days or longer than 54 days [2], the final sample comprised **1,554 cycles from 118 women** ($M = 13.2$ cycles per woman, $SD = 6.1$, range 5–45).

Raw cycle lengths ranged from 18 to 54 days ($M = 29.3$, $SD = 3.9$, $Mdn = 28.5$, skewness = 1.29). The distribution is approximately bell-shaped with a right tail, consistent

with the literature [2, 10]. Women ranged in age from 21 to 43 years ($M = 30.6$, $SD = 5.7$). The dataset originates from two charting cohorts ($n_0 = 92$ and $n_1 = 67$ women in the raw data prior to exclusions; $n_0 = 69$ and $n_1 = 49$ women in the filtered sample). All cycle lengths were incremented by 1 to conform to the halachic haflaga convention ($H_k = L_k + 1$), yielding $M = 30.3$, $SD = 3.9$.

The within-person variability structure is relevant to the Haflaga result: women who establish a Haflaga vessel ($n = 20$) have a mean within-person cycle-length SD of 1.54 days ($SD_{\text{between}} = 0.59$), compared to 3.17 days ($SD_{\text{between}} = 1.75$) for women who establish no vessel at all ($n = 35$). A Mann–Whitney test confirms this difference is highly significant ($U = 133$, $p < .001$, two-sided; Welch $t = -5.05$, $df = 45.5$, $p < .001$; 95% CI for mean difference: $[-2.29, -0.98]$ days; Cohen’s $d = -1.13$). This confirms that Haflaga selects the subgroup of women with the lowest observed cycle-length variability. Figure 1 shows the cycle length distribution and per-woman variability.

Figure 0. Menstrual cycle length distribution (filtered dataset)

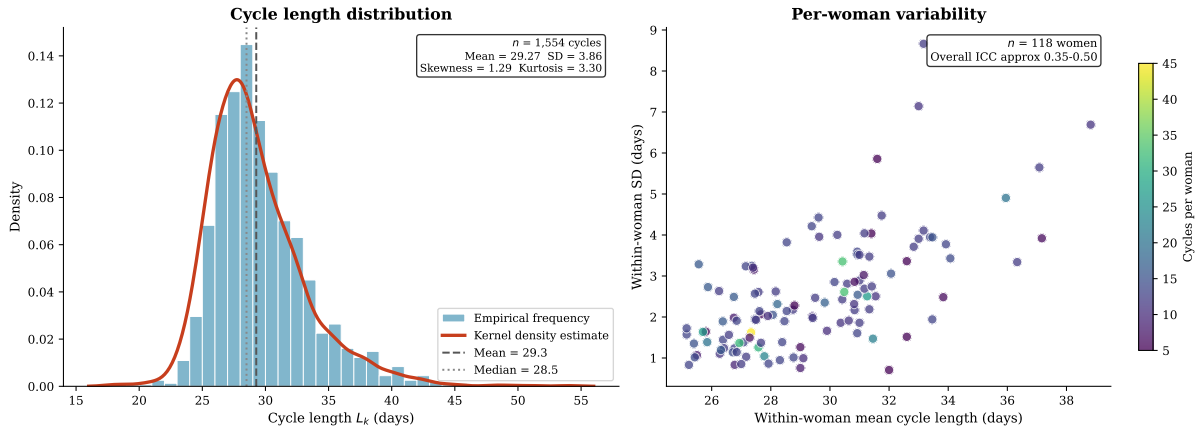


Figure 1: Cycle length distribution (filtered dataset). *Left:* histogram and kernel density estimate of raw cycle lengths L_k , with mean (dashed) and median (dotted) marked. The distribution is right-skewed (skewness = 1.29). *Right:* per-woman mean vs. within-person SD, coloured by cycle count. Women with more cycles tend to show lower variability, consistent with selection of naturally regular chartists.

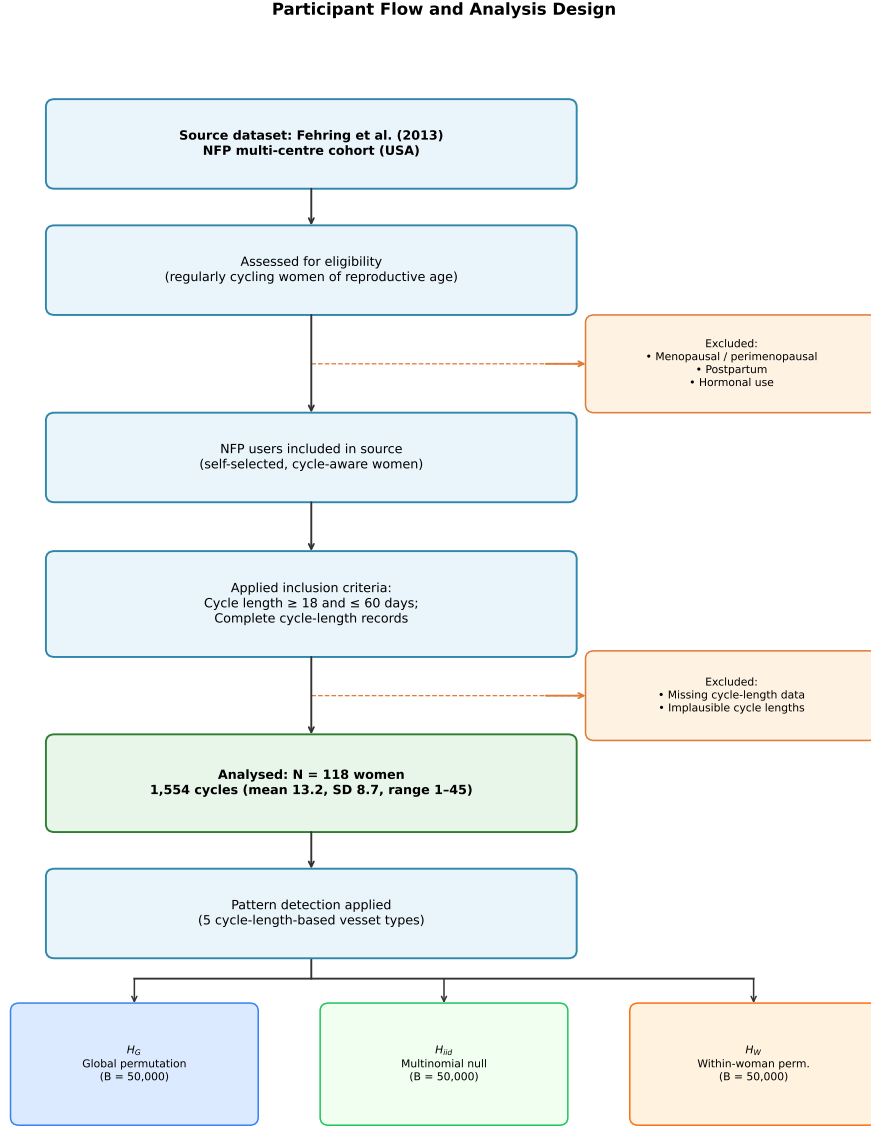


Figure 2: Participant flow and analysis design (CONSORT-style). The source dataset (Fehring et al., 2013) comprises data from a multi-centre NFP cohort. After applying inclusion criteria (cycle length 18–54 days; complete cycle-length records), the analysed dataset comprised 118 women contributing 1,554 cycles. Five pattern types were detected per woman. Three complementary null models were then applied: H_G (global permutation, $N = 50,000$ iterations), H_{iid} (multinomial null, $N = 50,000$ iterations), and H_W (within-woman permutation null, $N = 50,000$ iterations per run).

3.2 Pattern Detection

Pattern counting was performed within each woman’s sequence independently; sequences were never merged across women. Patient boundary pairs (s_i, f_i) were encoded explicitly and all counting loops were constrained to $[s_i, f_i]$. Extended runs of the same pattern were

counted as a single occurrence to avoid double-counting (e.g., four consecutive equal haflaga values constitute one Haflaga, not two). If a qualifying run was interrupted by a discordant cycle and a new qualifying run of the same type began later in the same woman’s sequence, that resumed run was counted as a separate event; the counts therefore reflect all distinct qualifying runs across each woman’s full history, not only the initial fixation event. All five functions were validated against manually annotated examples and are available at <https://github.com/dvirross/VesetVectorTest>.

One counting implication follows directly from the definitions: Week-Dilug requires $H_k = H_{k-1} = 30$ (two consecutive haflaga values equal to 30), whereas Haflaga-30 requires three consecutive haflaga values equal to 30. Therefore every Haflaga-30 sequence automatically generates a Week-Dilug event in its first two cycles:

$$\{\text{Haflaga-30 sequence}\} \subseteq \{\text{Week-Dilug sequence}\}.$$

This is set-theoretic inclusion, not merely statistical correlation, and it means the Haflaga and Week-Dilug counts are not independent: the 3 Haflaga sequences with $H = 30$ each contribute directly to the Week-Dilug count of 18.

3.3 Null Models

Three null models address distinct scientific questions:

H_G (global exchangeability): All 1,554 cycle lengths exchangeable across women. Tests whether patterns exceed population-level chance.

H_W (within-woman permutation): Each woman’s cycles permuted uniformly at random within her own sequence (without replacement). Tests whether temporal *ordering* creates excess patterns beyond individual regularity.

H_{iid} (multinomial i.i.d.): Each woman’s cycles drawn i.i.d. from $\hat{F}$. Stronger than H_G : replaces observed values, not just ordering.

H_G and H_{iid} are the global nulls (permutation and multinomial respectively); H_W is the within-woman null. All three were run for $N = 50,000$ iterations (seed 17).

Permutation test. — At each iteration, the $n = 1,554$ haflaga values are randomly permuted without replacement. This preserves the exact empirical marginal distribution and is equivalent to assuming all $n!$ orderings are equally probable under H_0 [5].

Multinomial null. — At each iteration, for each woman i with n_i cycles, a fresh sequence $\tilde{H}_{i,1}, \dots, \tilde{H}_{i,n_i} \stackrel{\text{i.i.d.}}{\sim} \hat{F}$ is drawn, where $\hat{F}$ is the empirical CDF of all 1,554 observed haflaga values. This null directly encodes the hypothesis that cycles are *independent* across time within each woman [16]. It differs from a naive global bootstrap (resampling with replacement from the pooled distribution, which is similar in spirit to the multinomial null) in that each woman draws from the *same* marginal $\hat{F}$, ignoring between-woman heterogeneity in cycle-length distributions. The within-woman permutation null (H_W) addresses this by conditioning each woman’s simulated sequence on her *own* observed cycle-length multiset. A within-woman bootstrap (sampling with replacement from each woman’s own empirical distribution) was evaluated as an alternative but rejected: because consecutive i.i.d. draws are independent whereas permutation draws are negatively correlated (using a value reduces its availability at the next position), the bootstrap inflates the expected count of repetition-based patterns such as Haflaga — from a null mean of 31.9 under permutation to 54.5 under the bootstrap (a 71% increase verified by $B = 50,000$ simulations). Permutation is therefore preferred as the exact conditional randomisation test for the stated exchangeability null; the bootstrap corresponds to a different iid-within-woman null, and the two procedures are not directly comparable as more or less conservative in general — their different expected counts reflect the different scientific questions they encode.

3.4 Statistical Inference

For each pattern j and null model m we computed: the observed count O_j ; the null mean $\bar{S}_j^{(m)}$ and SD $\sigma_j^{(m)}$; the z-score $z_j^{(m)} = (O_j - \bar{S}_j^{(m)})/\sigma_j^{(m)}$; and the one-sided empirical p-value $p_j^{(m)} = (b_j + 1)/(B + 1)$, where $b_j = \#\{S_{j,r}^{(m)} \geq O_j\}$ and $B = 50,000$, following the Monte Carlo convention of Davison & Hinkley [4].

The **joint multivariate test** uses the Euclidean distance of the observed count vector

from the null mean vector as the test statistic:

$$D^{(m)} = \|\mathbf{O} - \bar{\mathbf{S}}^{(m)}\|_2 = \sqrt{\sum_{j=1}^5 (O_j - \bar{S}_j^{(m)})^2}, \quad (1)$$

with p-value equal to the fraction of iterations where the simulated distance exceeded $D^{(m)}$. Unlike Bonferroni–Holm correction — which, while valid under arbitrary dependence, is conservative when tests are positively correlated — this joint permutation test operates on the full count vector simultaneously, respecting the dependence structure. Note that this is a single global test of the multivariate null, not an FWER procedure across the five component tests. Inspired by King et al. [9] but using distance rather than density as the test statistic, this approach is, to our knowledge, the first application of a distance-based joint permutation test to sequentially-defined pattern count vectors in longitudinal medical data. We adopt the **Mahalanobis distance** $D_M^2 = (\mathbf{O} - \boldsymbol{\mu})^\top \hat{\Sigma}^{-1} (\mathbf{O} - \boldsymbol{\mu})$ as the primary statistic, as it correctly standardises each component by its null variance and accounts for inter-component correlations. Non-diagonal terms in $\hat{\Sigma}$ are expected a priori: the set-theoretic containment of Week-Dilug within Haflaga-30 and the polynomial proximity of Dilug-in-Dilug to Haflaga (Section 2.2) imply that the five pattern counts are not marginally independent, so a diagonal (Euclidean) standardisation would misrepresent the dependence structure. $\hat{\Sigma}^{(m)}$ is estimated empirically from the $B = 50,000$ simulated count vectors under each null model, yielding a 5×5 sample covariance matrix. Condition numbers of 4.4 (permutation null) and 5.4 (multinomial null) confirm that the matrix is well-conditioned and the inversion is numerically stable. The Euclidean distance D (Eq. 1) is retained as a secondary statistic for geometric interpretability.

The **primary joint test** is the focused Mahalanobis test restricted to the three structurally dependent patterns — Haflaga, Week-Dilug, and Dilug-in-Dilug. The justification is purely structural, established before inspecting the data: $\text{Week-Dilug} \subseteq \text{Haflaga}$ by set-theoretic containment (Section 2.2), and Dilug-in-Dilug is the degree-2 polynomial neighbour of the degree-0 Haflaga in the hierarchy. These three therefore form a natural analytical unit, and the 3-pattern Mahalanobis statistic applies Eq. 1 to the subvector $(O_{\text{Haf}}, O_{\text{WD}}, O_{\text{DiD}})$ with a 3×3 covariance submatrix (condition number 2.6; well-conditioned). The **5-pattern test** over all

patterns is reported as a robustness check.

Sensitivity. — The joint test is sensitive primarily to the Haflaga component, which contributes the largest standardised excess ($z = 4.51$) and — after variance standardisation by the Mahalanobis metric — the largest weight. Under the global permutation null, the observed $D_M = 5.10$ exceeds the 99.9th percentile of the null distribution.

Pairwise **chi-square tests of independence** between pattern types were computed on the per-woman binary matrix (1 = woman established the pattern at least once). $\alpha = .05$ throughout.

Simulation pipeline. — Algorithm 1 provides a formal pseudocode description of the full simulation pipeline for reproducibility.

Algorithm 1 Full simulation pipeline

Require: Dataset $\mathcal{D} = \{(H_{i,1}, \dots, H_{i,n_i})\}_{i=1}^N$ (N women), patterns $j = 1, \dots, 5$, iterations B , seed

- 1: Compute observed counts $\mathbf{O} = (O_1, \dots, O_5)$ via pattern detection
- 2: **for** $r = 1$ to B **do**
- 3: // Global permutation null (H_G): shuffle all $\sum_i n_i$ haflaga values
- 4: $\mathbf{H}^{(G)} \leftarrow \text{Permute}(\bigcup_i \{H_{i,k}\})$; re-assign to women preserving n_i
- 5: $\mathbf{S}^{(G,r)} \leftarrow \text{Count}(\mathbf{H}^{(G)})$
- 6: // Multinomial null (H_{iid}): draw each $H_{i,k}$ iid from marginal distribution
- 7: $\mathbf{H}^{(M)} \leftarrow \text{Multinomial}(p_1, \dots, p_K; \text{total } \sum_i n_i)$
- 8: $\mathbf{S}^{(M,r)} \leftarrow \text{Count}(\mathbf{H}^{(M)})$
- 9: // Within-woman permutation null (H_W): shuffle within each woman
- 10: **for** $i = 1$ to N **do**
- 11: $\mathbf{H}_i^{(W)} \leftarrow \text{Permute}(H_{i,1}, \dots, H_{i,n_i})$ ▷ without replacement
- 12: **end for**
- 13: $\mathbf{S}^{(W,r)} \leftarrow \text{Count}(\mathbf{H}^{(W)})$
- 14: **end for**
- 15: Estimate $\hat{\Sigma}^{(m)} \leftarrow \text{SampleCov}(\{\mathbf{S}^{(m,r)}\}_{r=1}^B)$ for each $m \in \{G, M, W\}$
- 16: **for** $m \in \{G, M, W\}$ **do**
- 17: Null mean $\boldsymbol{\mu}^{(m)} \leftarrow B^{-1} \sum_r \mathbf{S}^{(m,r)}$
- 18: Mahalanobis statistic: $D_M^{(m)} \leftarrow \sqrt{(\mathbf{O} - \boldsymbol{\mu}^{(m)})^\top (\hat{\Sigma}^{(m)})^{-1} (\mathbf{O} - \boldsymbol{\mu}^{(m)})}$
- 19: $p_j^{(m)} \leftarrow (|\{r : S_j^{(m,r)} \geq O_j\}| + 1)/(B + 1)$ for each pattern j ▷ one-sided
- 20: $p_{DM}^{(m)} \leftarrow (|\{r : D_M^{(m,r)} \geq D_M^{(m)}\}| + 1)/(B + 1)$
- 21: **end for**

4. Results

4.1 Observed Pattern Counts

Observed counts were: Haflaga = 26, Dilug = 61, Week = 32, Week-Dilug = 18, Dilug-in-Dilug = 27. Permutation null means were: 11.30, 48.93, 27.00, 16.27, 27.17 respectively. Of

the 26 Haflaga sequences, 3 had value $H = 30$, each automatically constituting a Week-Dilug event (set-theoretic inclusion). Of the 27 Dilug-in-Dilug sequences, 16 had second difference $|d| = 1$ (near-Haflaga).

4.2 Per-Pattern Hypothesis Tests

Table 1 presents the full results under both null models. Figures 3 and 4 display the null distributions and effect sizes.

Table 1: Permutation and multinomial null-model results ($N = 50,000$ iterations). *Null Mean* and *Null SD* are from the permutation null. Bold p -values are significant at $\alpha = .05$ (one-sided).

Pattern	Obs.	Null Mean	Null SD	z	Perm. p	Multi. p
Haflaga	26	11.30	3.26	+4.51	< .001	< .001
Dilug	61	48.93	6.60	+1.83	.043	.043
Week	32	27.00	4.03	+1.24	.133	.188
Week-Dilug	18	16.27	3.28	+0.53	.345	.366
Dilug-in-Dilug	27	27.17	5.04	−0.03	.543	.544

Figure 1: Empirical null distributions (PMF) for each vessel pattern

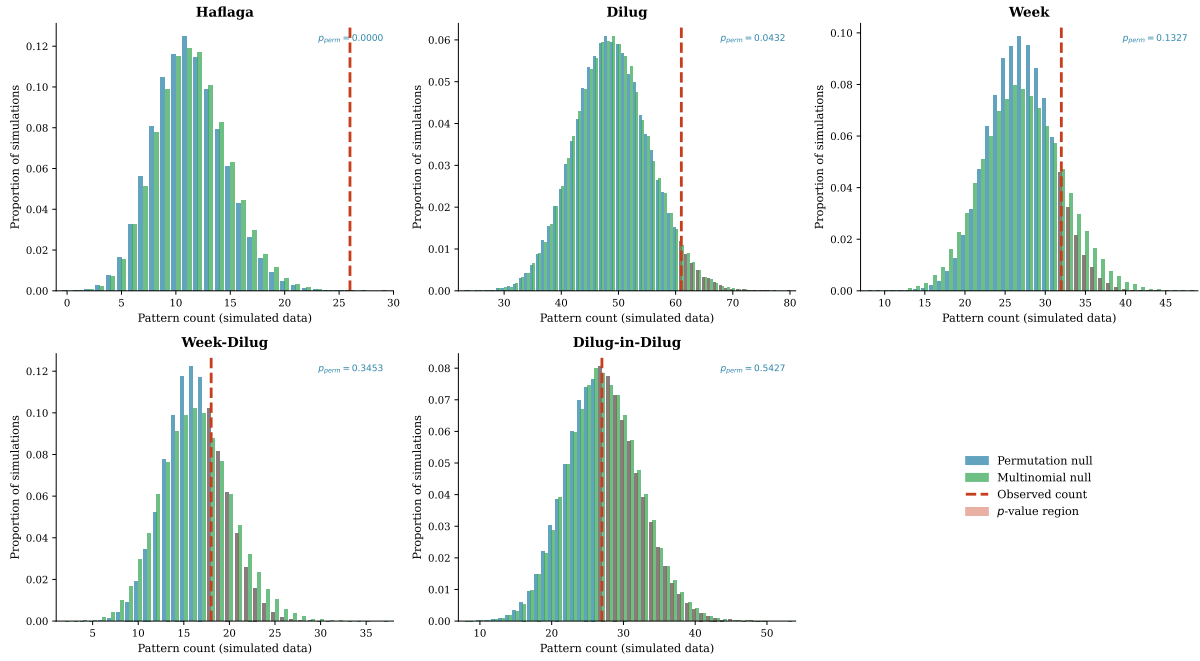
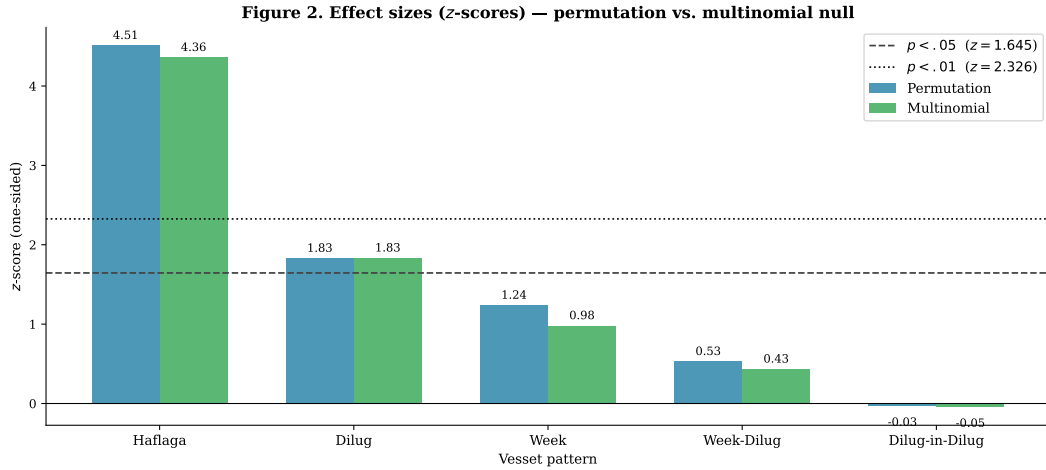


Figure 3: Null distributions (KDE) vs. observed counts for each vessel pattern under the permutation (blue solid) and multinomial (green dashed) null models. The vertical dotted red line marks the observed count; the shaded region is the one-sided p-value area. Haflaga and Dilug show clearly right-displaced observed values; Dilug-in-Dilug falls on its null mean.


 Figure 4: Effect sizes (z-scores) under permutation (blue) and multinomial (green) null models. Dashed and dotted lines mark $\alpha = .05$ ($z = 1.645$) and $\alpha = .01$ ($z = 2.326$) thresholds. Only Haflaga and Dilug exceed the $\alpha = .05$ threshold; both null models are in close agreement.

Haflaga showed the strongest effect: 26 occurrences vs. a null mean of 11.30, more than 4.5 standard deviations above chance ($z = 4.51$, exact Monte Carlo $p = 3/50,001 \approx .00006$). Crucially, under the permutation null only **2 of 50,000** simulations reached or exceeded 26 — the observed count ties the simulation maximum. Applying Holm–Bonferroni correction across

the five pattern tests, only Haflaga remains significant after FWER control (Holm-adjusted $p = .0003$). Dilug is nominally significant ($p = .043$) but does not survive Holm correction (adjusted $p = .173$) and should be regarded as an exploratory finding requiring replication. Week, Week-Dilug, and Dilug-in-Dilug are not significant.

4.3 Agreement Between Null Models

The permutation and multinomial null models are in close agreement across all five patterns (maximum difference in p-values: 0.055). Figure 5 shows the empirical CDFs, confirming the two nulls produce nearly identical distributions for each pattern. The close agreement is a numerical observation; whether and under what conditions the two procedures converge asymptotically depends on the specific sampling model and is not formally established here.

Figure 4: Empirical CDFs of null distributions for each vessel pattern

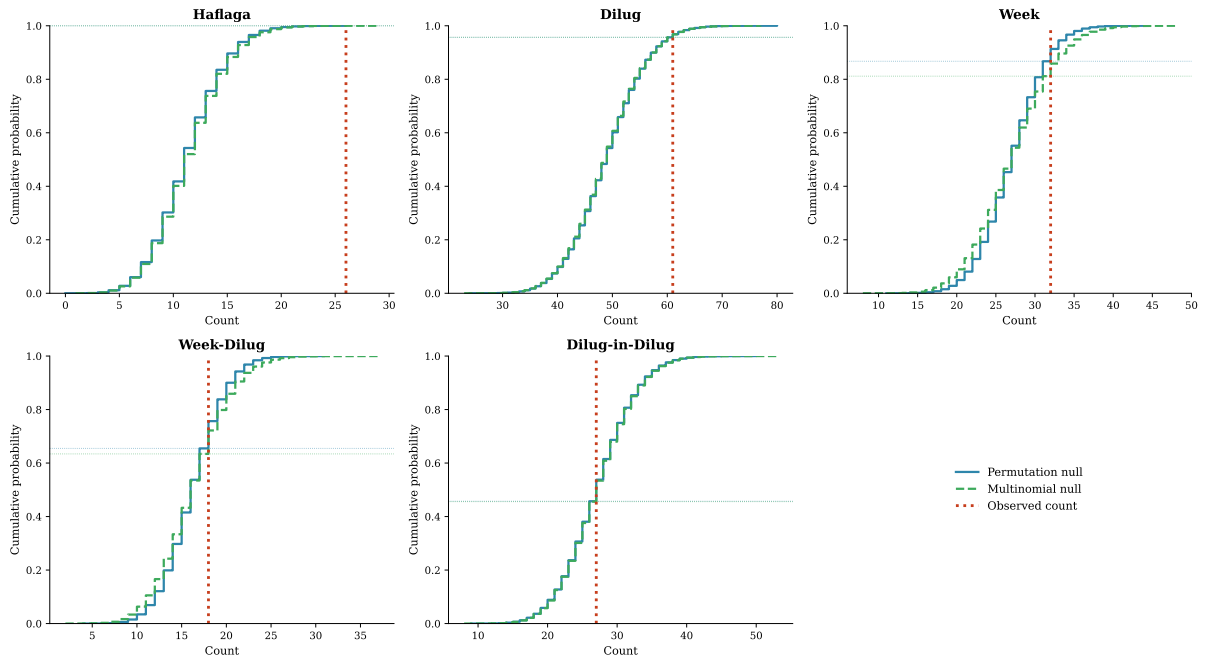


Figure 5: Empirical CDFs of null distributions for each pattern under permutation (blue solid) and multinomial (green dashed) nulls. The vertical dotted red line marks the observed count; horizontal dotted lines indicate $1 - p$. The close overlap confirms robustness to the choice of null model.

4.4 Within-Woman Permutation Analysis

The global permutation and multinomial null models permute or resample haflaga values *across* all 118 women, thereby removing between-woman heterogeneity in cycle regularity. As noted in Section 3.3, this means both global nulls test whether vessel patterns are more frequent than

expected if all women drew from the same population distribution — not whether patterns are more frequent than expected *given each woman’s own level of regularity*. These are distinct scientific questions.

To address the within-woman question, we implemented a **within-woman permutation** null (H_W): for each woman independently, her cycle-length sequence is randomly permuted (shuffled without replacement), preserving her individual cycle-length *multiset* while randomising the temporal ordering. This null encodes the hypothesis that the *temporal ordering* of cycles within a woman does not create patterns beyond what her own cycle-length values (taken in random order) would predict.

Implementation. — For each of the 118 women independently, her cycle-length sequence is permuted uniformly at random (without replacement). All women are included with no exclusion threshold. The aggregate null is generated by summing the permuted count vectors across all 118 women, producing $B = 50,000$ realisations of the joint count vector.

Results. — Table 2 presents the results. Under the within-woman null, *no pattern is individually significant*. All five observed counts fall *below* their within-woman null means (all $z < 0$). The most negative z-score belongs to Dilug-in-Dilug ($z = -2.07$, $p = .988$); because all tests are one-sided upper-tail, this large negative z indicates that the count is well below the null mean — not that it is unexpectedly low — and should not be interpreted as evidence of underrepresentation. The joint test is also non-significant: Mahalanobis $D_M = 2.95$ ($p = .121$), Euclidean $D_E = 15.2$ ($p = .145$). Tables 2 and 4 both use the primary 50,000-run simulation (seed 17) to ensure full numerical consistency of μ_W , σ_W , and z_{obs} across all results.

Table 2: Within-woman permutation null (H_W) results ($N = 50,000$; all $n = 118$ women; each woman’s sequence permuted uniformly at random without replacement). All observed counts fall below their within-woman null means. p -values use the $(b + 1)/(B + 1)$ convention.

Pattern	Obs.	Null Mean	Null SD	z	p
Haflaga	26	31.85	4.68	−1.25	.916
Dilug	61	67.07	7.36	−0.83	.813
Week	32	33.81	3.88	−0.47	.722
Week-Dilug	18	20.94	3.22	−0.91	.857
Dilug-in-Dilug	27	39.14	5.86	−2.07	.988
Joint: $D_M = 2.95$ ($p = .121$); Euclidean $D = 15.2$ ($p = .145$)					

Figure 6 illustrates the contrast between the global and within-woman nulls for each pattern. The shift in the null distribution is striking: for Haflaga, the global null is centred around 10, while the within-woman null is centred around 27 — far above the observed count of 26 (all 118 women).

Figure 5. Three null models — all 118 women
 Global permutation (blue), global multinomial (green), within-woman stratified permutation (orange)
 Per-pattern: empirical PMFs; joint panel: Mahalanobis D_M

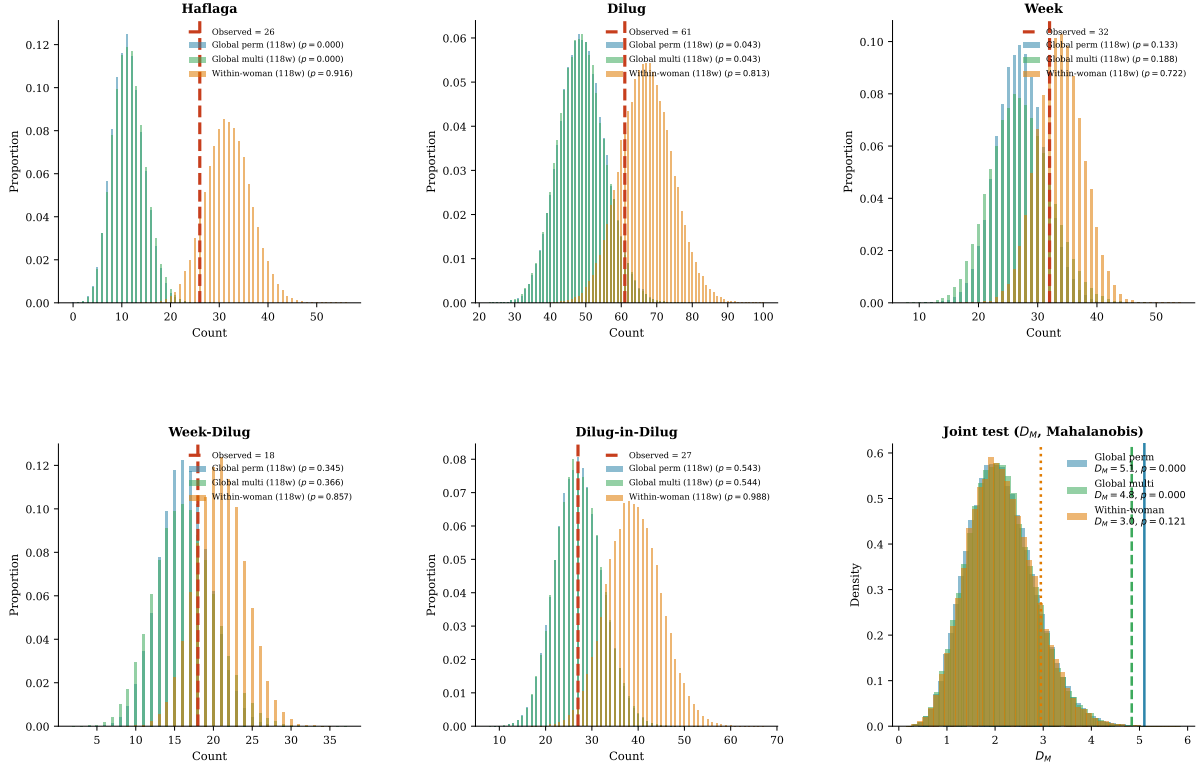


Figure 6: Null distributions (H_G , H_{iid} , H_W) for each pattern under three null models: global permutation (blue, 118 women), global multinomial (green, 118 women), and within-woman permutation (orange, 118 women). Dashed vertical line: observed count (118 women). The joint panel shows Mahalanobis D_M histograms; both global nulls are highly significant ($p < .001$) while the within-woman null is non-significant ($D_M = 2.95$, $p = .121$).

Interpretation. — The two nulls answer different scientific questions and are both valid:

- The **global null** tests: *are vessel patterns more frequent than expected if all cycle lengths were exchangeable across all women?* The answer is yes for Haflaga ($p < .001$ after Holm correction), reflecting that vessel patterns are concentrated among women who are intrinsically more regular.
- The **within-woman null** tests: *given each woman's own level of cycle regularity, are the temporal arrangements unusually structured?* The answer is no — under the within-woman exchangeability null, this dataset provides no detectable evidence that temporal ordering adds structure beyond each woman's own cycle-length distribution.

Together, these results suggest that halachic vessel rules — particularly Haflaga — constitute an

empirical codification of individual menstrual regularity: the rules correctly identify the subset of women whose cycles are intrinsically stable, and formalise that stability as an anticipatory pattern. Whether this reflects deliberate halachic wisdom or fortuitous alignment with biology is beyond the scope of the present analysis. A companion study examining the practical forecasting efficiency of halachic vessel rules — how accurately they predict the next menstruation onset, and at what cost in unnecessary abstinence days — is in preparation [12].

4.5 Joint Multivariate Test

Primary test: focused 3-pattern Mahalanobis. — The primary joint test operates on the structurally dependent subvector (Haflaga, Week-Dilug, Dilug-in-Dilug), justified on analytical grounds prior to data inspection (Section 3.4). The 3-pattern Mahalanobis statistic was $D_M = 4.52$ ($p < .001$) under the permutation null and $D_M = 4.36$ ($p < .001$) under the multinomial null. Figures 9 and 10 illustrate the displacement of the observed 3-pattern vector from the respective null clouds, using these three patterns as the coordinate axes.

Robustness: 5-pattern Mahalanobis. — Extending the joint test to all five patterns yields $D_M = 5.10$ ($p < .001$, permutation) and $D_M = 4.84$ ($p < .001$, multinomial). The Euclidean 5-pattern distance was $D = 19.74$ ($p = .009$, permutation; $p = .014$, multinomial). Figure 7 shows the 5-pattern Euclidean null distributions. The agreement between all four variants confirms that the primary result is robust to the choice of pattern subset and distance metric.

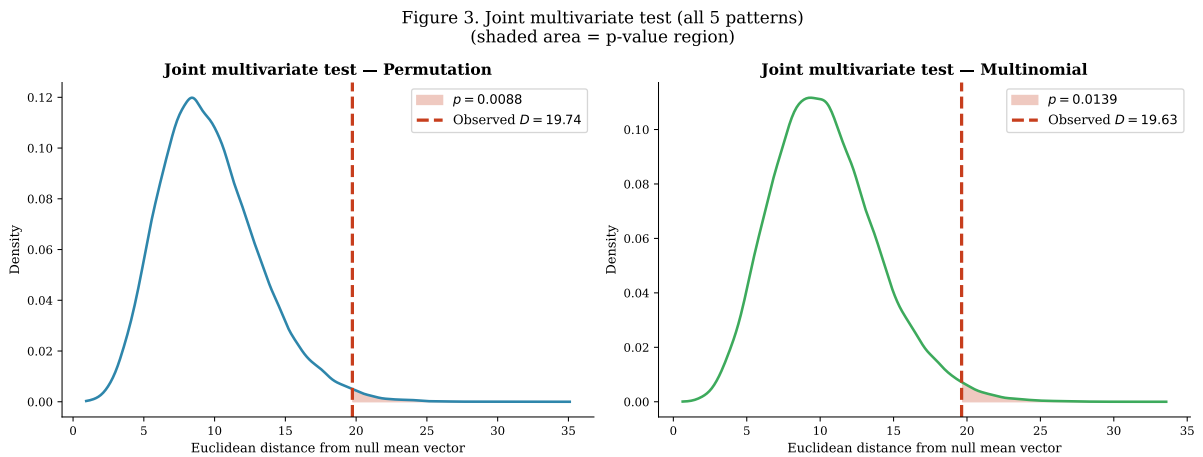


Figure 7: Robustness check: null distribution of 5-pattern Euclidean distances $D = \|\mathbf{S} - \tilde{\mathbf{S}}\|_2$ under the permutation (left) and multinomial (right) nulls. The dashed line marks the observed $D = 19.74$; the shaded region is the p-value area. Both models yield $p < .01$.

4.6 Pairwise Dependence Tests and Focused Joint Test

Table 3 presents the pairwise dependence tests. These analyses are **exploratory**: the tests were not pre-specified, multiple pairs are tested simultaneously, and the primary scientific contribution of these comparisons is to confirm the mathematical structure derived in Section 2.2 rather than to establish new independent findings. Each pair is tested with both Fisher’s exact test and a randomisation-based permutation test (50,000 random shuffles of the binary column; no distributional assumption), addressing the concern that small expected cell counts (< 5) may inflate asymptotic χ^2 p -values and the potential confound that women with more cycles have more opportunity to establish multiple patterns (the permutation test preserves each woman’s column marginal, eliminating this confound by design). The two methods agree closely across all pairs. Two pairs are statistically significant: Haflaga $\times$ Week-Dilug ($\chi^2 = 7.26$, $p_{\text{perm}} = .013$) and Haflaga $\times$ Dilug-in-Dilug ($\chi^2 = 9.99$, $p_{\text{perm}} = .004$), both with positive direction (co-occurrence substantially exceeds chance expectation). These findings are consistent with — and explained by — the mathematical relationships derived in Section 2.2. Figure 8 visualises the full dependence matrix.

Table 3: Pairwise dependence tests between vessel types (per-woman binary, $N = 118$). Significant pairs in bold. χ^2 : Pearson chi-square statistic; p_F : Fisher’s exact p -value (two-sided; used for all pairs due to small expected cell counts); p_{perm} : randomisation-based p -value (50,000 permutations of the binary column; exact under the null that co-occurrence arises by chance, independent of asymptotic distributional assumptions). Direction: $+$ = co-occur more than expected; $-$ = less than expected. The two methods agree closely, confirming that significance is not an artefact of asymptotic approximations. Notably, the Haflaga $\times$ Dilug-in-Dilug permutation p -value (0.004) falls below 0.01, providing strong evidence against the null that co-occurrence arises by chance even under non-asymptotic conditions.

Pair	χ^2	p_F	p_{perm}	Dir.	Obs(1,1)	Exp(1,1)
Haflaga $\times$ Dilug	0.87	.455	.459	+	10	8.1
Haflaga $\times$ Week	0.89	.379	.381	+	6	4.4
Haflaga $\times$ Week-Dilug	7.26	.014	.013	+	7	3.1
Haflaga $\times$ Dilug-in-Dilug	9.99	.004	.004	+	9	3.9
Dilug $\times$ Week	2.40	.174	.173	+	14	10.6
Dilug $\times$ Week-Dilug	1.46	.300	.300	-	5	7.3
Dilug $\times$ Dilug-in-Dilug	0.09	.815	.817	+	10	9.4
Week $\times$ Week-Dilug	0.41	.542	.546	+	5	4.0
Week $\times$ Dilug-in-Dilug	1.17	.276	.398	+	7	5.1
Week-Dilug $\times$ Dilug-in-Dilug	0.93	.342	.518	+	5	3.5

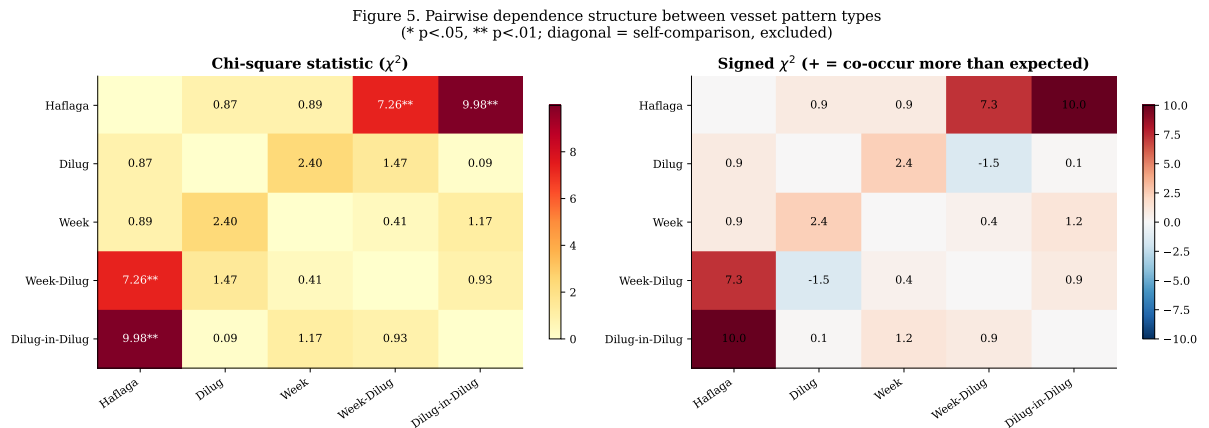


Figure 8: Pairwise dependence structure between vessel pattern types. *Left:* chi-square statistic; *right:* signed χ^2 (positive = co-occurrence exceeds expectation; negative = below expectation). The two significant pairs (Haflaga $\times$ Week-Dilug and Haflaga $\times$ Dilug-in-Dilug, marked **) are both positive, consistent with the mathematical derivations in Section 2.2.

The primary 3-pattern Mahalanobis joint test is reported in Section 4.5. Figures 9 and 10 provide complementary geometric illustrations of the displacement of the observed count vector in the (Haflaga, Week-Dilug, Dilug-in-Dilug) subspace, under the permutation and multinomial nulls respectively. Both use these three patterns as coordinate axes; the remaining two patterns (Dilug, Week) are encoded by marker size and colour.

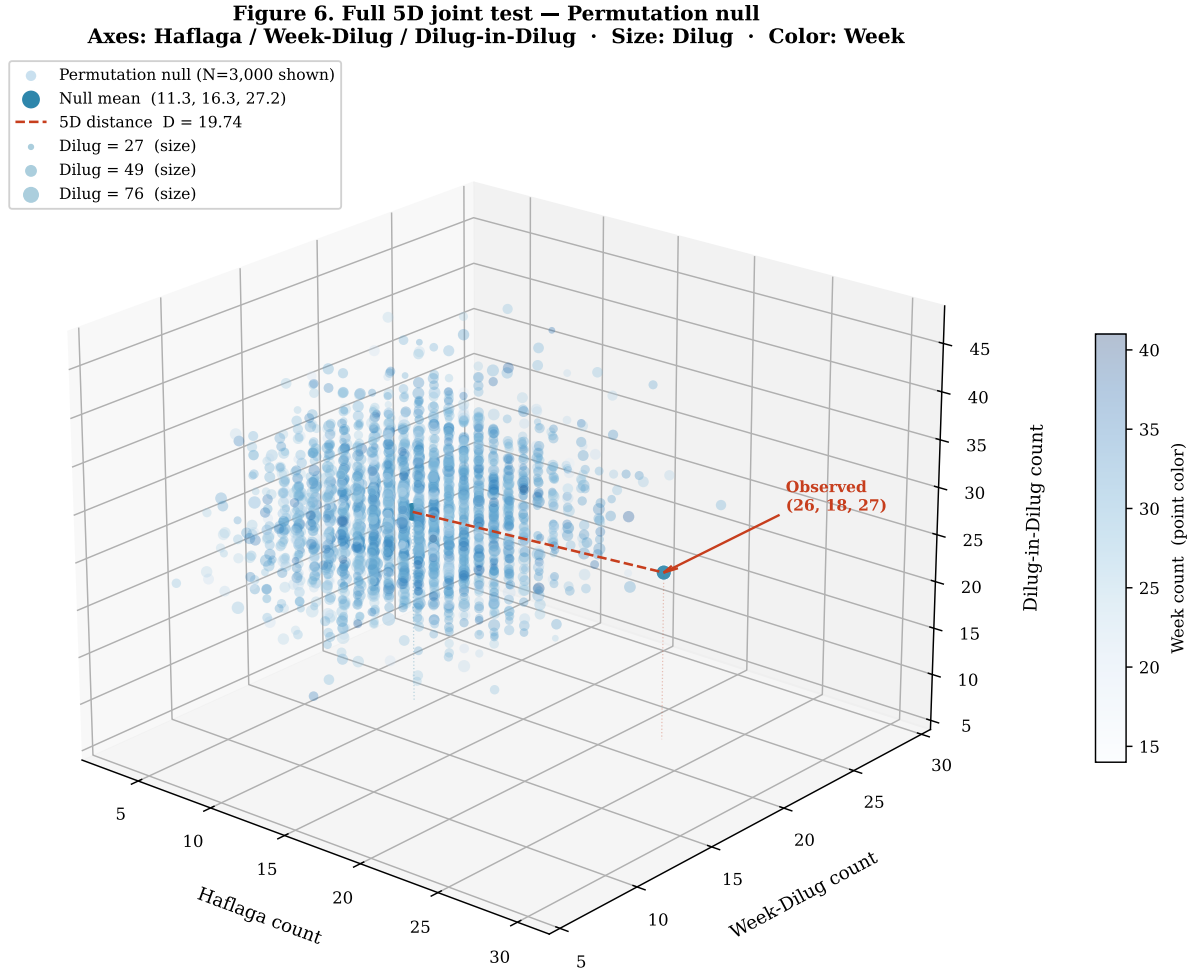


Figure 9: Geometric illustration of the focused 3-pattern joint test under the **permutation** null. Axes: Haflaga, Week-Dilug, Dilug-in-Dilug — the three structurally dependent patterns. Marker size encodes Dilug count; marker shape encodes Week count. The filled circle marks the null mean; the arrow and label identify the observed vector (26, 18, 27). The observed vector lies at the periphery of the null cloud, displaced primarily along the Haflaga axis. Primary result: $D_M = 4.52, p < .001$.

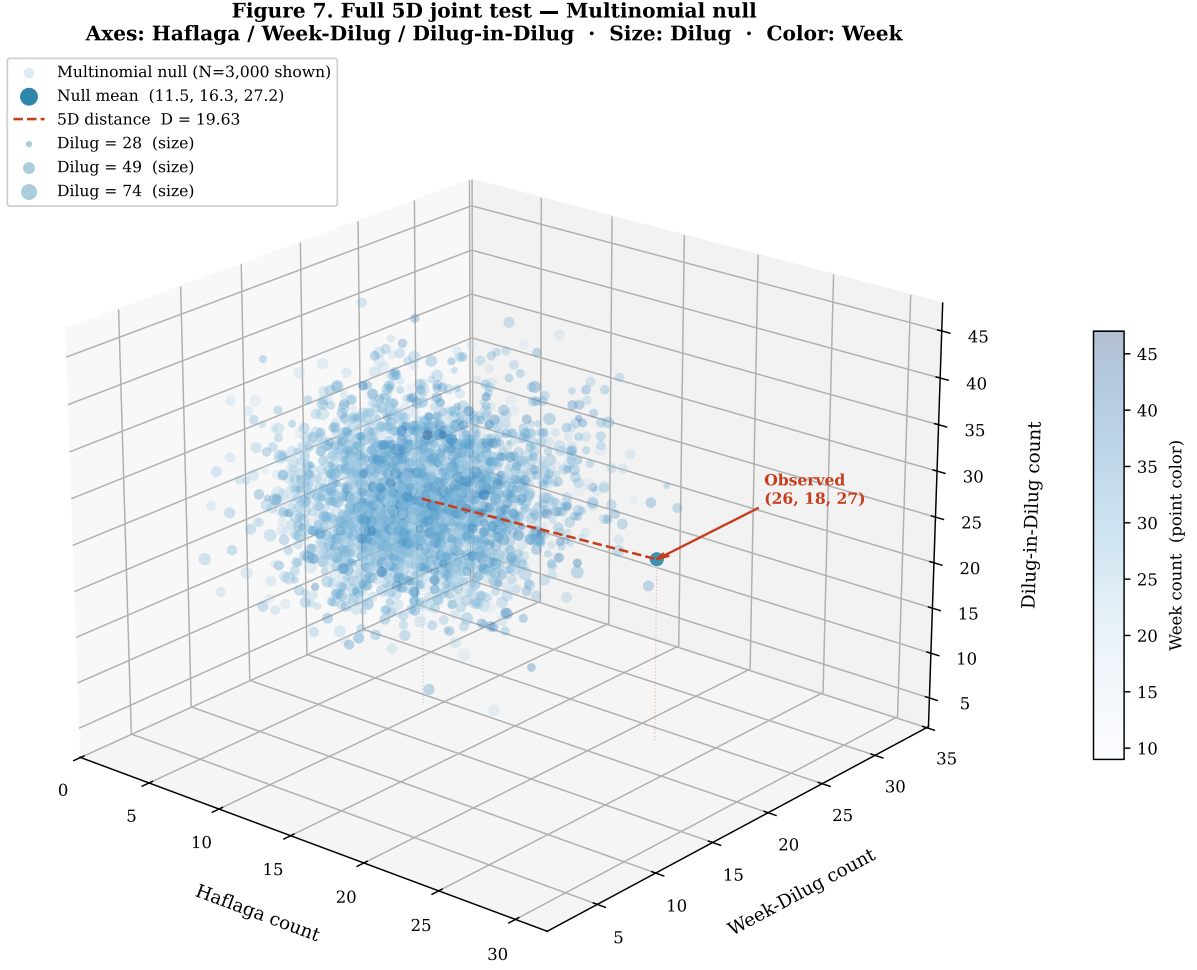


Figure 10: Geometric illustration of the focused 3-pattern joint test under the **multinomial** null. Encoding identical to Figure 9. Primary result: $D_M = 4.36$, $p < .001$. The close agreement between the two null clouds confirms that the result is robust to the choice of null model.

5. Discussion

This study provides the first formal statistical test of whether halachic vessel patterns are overrepresented in empirical menstrual cycle data. The central result is from the within-woman permutation null (H_W): no pattern is individually significant (all $p \geq .722$; joint $D_M = 2.95$, $p = .121$), and all observed counts fall below their within-woman null means. This dataset provides no detectable evidence, under within-woman exchangeability, that temporal ordering within each woman's cycle sequence creates vessel patterns beyond her own cycle-length distribution.

Under the global permutation null, which pools cycles across all 118 women, *Haflaga* is strongly overrepresented ($z = 4.51$, $p < .001$ after Holm correction) and *Dilug* is nominally

overrepresented ($p = .043$, does not survive Holm correction). The joint Mahalanobis test under the global null confirms significant displacement ($D_M = 5.10$, $p < .001$), driven primarily by Haflaga. The global overrepresentation reflects concentration of patterns among women who are intrinsically more regular — a population-level heterogeneity effect, not evidence of within-woman temporal structure.

5.1 Haflaga

The Haflaga finding is the most striking result. With only 2 of 50,000 permutation simulations reaching or exceeding the observed count of 26 (the simulation ceiling), the result is highly significant under the global null. It is biologically interpretable: Haflaga detects the pattern expected in women with lower within-person cycle-length variability. The present data confirm this association: women who establish Haflaga have within-person SDs of 1.54 days ($n = 20$) vs. 3.17 days for women who establish no vessel at all ($n = 35$; Mann–Whitney $p < .001$, Cohen’s $d = -1.13$; see Section 3.1 for full inference). These findings are consistent with the interpretation that Haflaga is empirically characteristic of women with lower observed within-woman cycle-length variability.

5.2 Dilug

The Dilug finding ($p = .043$ under the global null) is nominally significant but does not survive Holm–Bonferroni correction (adjusted $p = .173$) and should be treated as a hypothesis-generating result requiring pre-registered replication in larger samples. The direction is consistent with arithmetic progressions in cycle length arising from gradual biological trends (e.g., gradual lengthening or shortening associated with hormonal change, ageing, or weight change), but the evidence is insufficient to provide evidence for a population-level excess beyond chance. These biological explanations are speculative and await independent confirmation.

5.3 Mathematical Dependence Structure as a Unifying Observation

A distinctive feature of this study is that the observed statistical dependences between pattern types are not post-hoc empirical discoveries but *mathematical necessities* derivable from the pattern definitions alone. This is, to our knowledge, an observation that has not previously been

stated explicitly in the halachic or statistical literature.

The finite-difference hierarchy. — The five vessel patterns are naturally unified by the language of finite differences. Recall that the n -th forward difference operator Δ^n is defined recursively by $\Delta^0 H_k = H_k$ and $\Delta^n H_k = \Delta^{n-1} H_{k+1} - \Delta^{n-1} H_k$. Each vessel pattern corresponds to a condition on a specific order of Δ :

Pattern	Condition	Order
Week-Dilug	$H_k = 30$ (constant at 30)	$\Delta^0 H_k = 30$
Week	$H_k = c \in \mathcal{W}$ (constant, anchored)	$\Delta^0 H_k = c$
Haflaga	$H_k = c$ (constant, any c)	$\Delta^0 H_k = c$
Dilug	$\Delta^1 H_k = d \neq 0$ (constant non-zero first difference)	
Dilug-in-Dilug	$\Delta^2 H_k = d \neq 0$ (constant non-zero second difference)	

This ordering mirrors the classical Taylor expansion: a constant function is a degree-0 polynomial, a linear function is degree-1, and a quadratic is degree-2. Haflaga, Week, and Week-Dilug detect degree-0 sequences (with progressively tighter constraints on which constant value is admissible); Dilug detects degree-1 sequences ($H_k = a + bk$, $b \neq 0$); and Dilug-in-Dilug detects degree-2 sequences ($H_k = a + bk + ck^2$, $c \neq 0$). Crucially, a degree-2 sequence with a very small leading coefficient c is nearly linear, and a degree-1 sequence with a very small slope b is nearly constant — so the hierarchy does not impose rigid boundaries; rather, it describes a spectrum of regularity along which the patterns sit at adjacent levels.

This graded structure generates three mathematically necessary relationships between the patterns.

Haflaga and Week-Dilug: containment. — Week-Dilug requires two consecutive haflaga values both equal to 30, i.e. $\Delta^0 H_{k-1} = \Delta^0 H_k = 30$. Haflaga-30 requires three consecutive haflaga values all equal to 30, i.e. $\Delta^0 H_{k-2} = \Delta^0 H_{k-1} = \Delta^0 H_k = 30$, which strictly contains the Week-Dilug condition. Algebraically,

$$\{H_{k-2} = H_{k-1} = H_k = 30\} \implies \{H_{k-1} = H_k = 30\},$$

so every Haflaga-30 sequence necessarily generates a Week-Dilug event at the same position. This is not a probabilistic statement but a deterministic logical implication: any haflaga sequence that satisfies the Haflaga-30 condition automatically satisfies the Week-Dilug condition at its first two elements. In the data, 3 of the 26 Haflaga sequences have value $H = 30$ and therefore each count once towards the Week-Dilug total as a direct algebraic consequence.

Haflaga and Dilug-in-Dilug: limiting convergence. — A Dilug-in-Dilug sequence with second difference d has first differences $\Delta^1 H_k = b, b + d, b + 2d, \dots$, so the haflaga values are

$$H_k = H_0 + bk + \frac{d}{2}k(k-1).$$

As $d \rightarrow 0$, this reduces to $H_k = H_0 + bk$ (a Dilug), and as both $d \rightarrow 0$ and $b \rightarrow 0$, it reduces to $H_k = H_0$ (a Haflaga). More precisely, for a run of length n starting at H_0 , the maximum deviation from the constant sequence $\mathbf{c} = H_0$ is bounded by

$$\|H - \mathbf{c}\|_\infty \leq |b|n + \frac{|d|}{2}n^2.$$

A DiD sequence with $|d| = 1$ and a short run ($n = 4$, the minimum) therefore deviates from the nearest constant by at most $4|b| + 8$ days — for small $|b|$, this is a sequence that is barely distinguishable from a Haflaga. In the data, 16 of the 27 observed DiD sequences have $|d| = 1$, the minimum non-zero value; for these, the haflaga values change by only 1 day per cycle relative to linear trend. The women who sustain a DiD pattern with $|d| = 1$ are the same women whose cycles are regular enough to sustain a near-constant sequence across multiple cycles — women with a similar degree of cycle regularity to Haflaga women. This explains the strong positive dependence ($\chi^2 = 9.99$, $p_{\text{perm}} = .004$): the finite-difference hierarchy places Haflaga and DiD at adjacent levels of polynomial order, and the data show that real women tend to cluster near the boundary between them.

Dilug and Week-Dilug: mutual exclusion. — Dilug requires $\Delta^1 H_k = H_k - H_{k-1} = d \neq 0$; Week-Dilug requires $\Delta^1 H_k = H_k - H_{k-1} = 0$. Within any single cycle window $(k-1, k)$,

these conditions are contradictory:

$$\Delta^1 H_k \neq 0 \quad \text{and} \quad \Delta^1 H_k = 0$$

cannot both hold. A cycle pair that satisfies Week-Dilug ($\Delta^1 H = 0$) is therefore excluded from contributing to a Dilug run ($\Delta^1 H \neq 0$), and vice versa. This mutual exclusion applies strictly at the *sequence window level*: a single cycle pair cannot contribute to both patterns simultaneously. A woman can, however, exhibit Dilug in one portion of her cycle history and Week-Dilug in another, so window-level mutual exclusion does not imply a deterministic negative association between the woman-level ever/never indicators. The observed direction is negative (5 co-occurrences vs. 7.3 expected under independence, $\chi^2 = 1.46$, $p_{\text{perm}} = .300$), which is consistent with the algebraic prediction but is not a statistical consequence of it.

Implication for inference. — The three relationships derived above — containment, limiting convergence, and mutual exclusion — show that the five pattern counts are *structurally* non-independent: their joint distribution is constrained by the algebra of finite differences, not by coincidence. Applying Bonferroni–Holm correction controls the family-wise error rate (FWER) under arbitrary dependence and does not require independence. However, it is known to be conservative when tests are positively correlated, as is the case here. The joint permutation test evaluates the count vector as a single entity, avoiding any multiple-comparison correction and sidestepping the conservatism of Holm–Bonferroni under positive dependence. The joint multivariate test of Section 3.4 operates directly on the count vector, without any independence assumption, thereby respecting the dependence structure that the finite-difference hierarchy imposes.

5.4 The Joint Multivariate Test

The joint test (Eq. 1) represents a methodological contribution [16]. We adopt the **Mahalanobis distance** as the primary test statistic after confirming that the null covariance matrices are well-conditioned (condition numbers 4.4–5.4; all pairwise correlations $|r| < 0.16$, so inversion is numerically stable). Because $\hat{\Sigma}$ is estimated entirely from the Monte Carlo null replicates (not from the observed data) and all pairwise correlations are small, a robust sandwich-type

estimator was not required; the simulation-based estimate is consistent by construction. The Mahalanobis statistic is $D_M^2 = (\mathbf{O} - \boldsymbol{\mu})^\top \hat{\Sigma}^{-1} (\mathbf{O} - \boldsymbol{\mu})$, which standardises deviations by the full null covariance matrix $\hat{\Sigma}$, including off-diagonal cross-product terms. Because $\hat{\Sigma}$ is estimated as approximately (though not exactly) diagonal here (all pairwise $|r| < 0.16$), the statistic is numerically close to $\sum_j (O_j - \mu_j)^2 / \sigma_j^2$ in this application, giving components with smaller null variance proportionally greater weight. Haflaga has the smallest null SD of the five (3.26 vs. 6.60 for Dilug), so its deviation of $(26 - 11.30) = 14.7$ units yields the largest standardised contribution. Condition numbers of 4.4 (permutation null) and 5.4 (multinomial null) confirm that inversion of $\hat{\Sigma}$ is numerically stable.

The primary result — the focused 3-pattern Mahalanobis test — yields $D_M = 4.52$ ($p < .001$) under the permutation null and $D_M = 4.36$ ($p < .001$) under the multinomial null. The 5-pattern robustness check yields $D_M = 5.10$ ($p < .001$, permutation) and $D_M = 4.84$ ($p < .001$, multinomial). All four variants yield highly significant results. The 5-pattern values yield approximately 100-fold smaller p-values than the Euclidean counterparts ($p = .009$ and $.014$), because Haflaga receives its correctly up-weighted contribution via the Mahalanobis metric.

Simulation algorithm. — The covariance matrix $\hat{\Sigma}$ is estimated once per null model from the same $B = 50,000$ simulated count vectors. The same fixed $\hat{\mu}$ and $\hat{\Sigma}$ are then used to evaluate the Mahalanobis statistic for every null replicate (a fully Monte Carlo reference distribution, not a leave-one-out procedure). The observed count vector is not included in $\hat{\mu}$ or $\hat{\Sigma}$. This procedure is computationally equivalent for all three null models and yields a purely empirical p-value without parametric assumptions.

This approach is inspired by King et al. [9] and related to energy-distance tests [18] and multivariate permutation tests [11], but differs in using Mahalanobis distance from the simulated null mean — requiring no density estimation, pairwise distances, or asymptotic approximations. The p-value is purely empirical. A formal theoretical treatment is planned in future work.

Relationship to Hotelling’s T^2 . — The permutation-based Mahalanobis test is related to, but more general than, Hotelling’s T^2 . Under a Gaussian approximation (i.e., assuming the count vector is approximately multivariate normal, as holds asymptotically via the central

limit theorem for large counts), D_M^2 is asymptotically $\chi^2(5)$, which is the same distributional assumption underlying T^2 . However, with observed counts as small as 18 (Week-Dilug), this approximation may not hold reliably — motivating the permutation reference distribution, which replaces the parametric assumption with an empirical one and makes the test fully distribution-free. Hotelling’s T^2 is an exact test under multivariate Gaussian sampling but requires this distributional assumption for validity. Moreover, the Bonferroni–Holm correction applied to the five marginal tests is provably valid but conservative under positive correlation (as holds here via the set-theoretic containment and polynomial proximity relationships), while the joint Mahalanobis test exploits the known positive correlation structure by pooling the five components simultaneously. For both reasons — distribution-free inference and proper treatment of dependence — the permutation-based Mahalanobis test is the preferred approach for this application.

5.5 Non-Significant Patterns

Week, Week-Dilug, and Dilug-in-Dilug were not individually significant. These results do not invalidate these halachic rules, which are established on legal-theological grounds [14]. The non-significance of Dilug-in-Dilug is particularly notable given its large z-score at the sequence level in the joint test: it is the dependence with Haflaga (via the polynomial hierarchy) that elevates its collective contribution to D , even when its individual count is exactly at the null mean. This illustrates why joint and individual tests can differ.

For the Week pattern, it is worth noting that the weekly anchor set $\mathcal{W} = \{22, 29, 36, 43, 50\}$ (in haflaga units) corresponds to cycle lengths $\{21, 28, 35, 42, 49\}$. Since cycles of 35 days or longer are well outside the normal range for the Fehring dataset (mean $L = 29.3$, $SD = 3.9$), nearly all Week patterns in this sample arise from haflaga values of 22 (cycle length 21) or 29 (cycle length 28). The 28-day cycle is the mode of the dataset and the most biologically common length, so the Week pattern in this dataset effectively captures women with near-28-day cycles who also show the requisite two-consecutive-cycle consistency.

5.6 Clinical and Halachic Implications

The findings carry several implications, all of which should be understood as hypotheses for future study rather than clinical guidance: none of the outcomes discussed below is directly evaluated in the present analysis.

Fertility awareness. — Women with established Haflaga represent the most cycle-regular subgroup in this dataset (within-person SD: 1.54 days vs. 3.17 days for non-vesset women). For such women, backward estimation of the fertile window (approximately 14 days prior to anticipated onset) is most reliable, making Haflaga a candidate marker for calendar-based fertility planning [1]. Whether this marker improves on statistical cycle-length prediction methods will be examined in the companion forecasting study [12].

Halachic family purity compliance. — For observant Jewish women, the vesset system governs the timing of ritual abstinence (*taharat hamishpacha*). The finding that Haflaga is statistically detectable as a regularity pattern is consistent with its role in halachic anticipation, while the non-significance under the within-woman null is consistent with the halachic position that vesset rules codify observed regularity rather than impose structure beyond it. Whether these empirical findings bear on the legal-theological validity of vesset rules is a question outside the scope of this analysis.

Cycle regularity as a health indicator. — Menstrual cycle irregularity is associated with decreased fertility and elevated risk of chronic conditions [1]. The finding that Haflaga marks women with low within-person cycle-length variability ($SD < 1.6$ days) suggests it could function as a simple regularity indicator — a clinical hypothesis to be evaluated in datasets with longitudinal clinical measurements and defined clinical endpoints. Changes in vesset status (loss or gain of Haflaga) may also serve as a signal of disruption to baseline regularity, warranting prospective investigation.

5.7 Scope and Generalizability

Before discussing specific limitations, we emphasise that all conclusions are restricted to this specific dataset of non-Jewish NFP users. The results should not be generalised to the general

population or to observant Jewish women without replication. The NFP dataset represents a self-selected group of cycle-aware women, likely more regular than the general population, which may inflate pattern counts. Any inference about the halachic validity of vessel rules in the Jewish context requires datasets collected from the target population. Such validation within an observant Jewish demographic would directly bridge the statistical findings to their halachic application context.

5.8 Limitations

Several further limitations of the dataset and methodology are acknowledged, largely following the systematic assessment in [16].

Relabelling invariance ($H_k = L_k + 1$). — The haflaga convention $H_k = L_k + 1$ is a relabelling that shifts all values by one day. As a sensitivity check, we verified that all pattern counts are identical whether computed on H_k or directly on L_k *with the anchor adjusted accordingly* (Week-Dilug at $L = 29$ instead of $H = 30$; emphasis added). This anchor adjustment is critical: computing on L_k with the unadjusted anchor ($L = 30$) yields only 12 Week-Dilug events instead of the correct 18. Provided the anchor is shifted by the same -1 that maps H_k to L_k , the counts are unchanged (Haflaga: 26, Week-Dilug: 18 under both conventions), confirming that the $+1$ transformation is not a substantive modelling choice but a conventional relabelling consistent with halachic convention [14, 16].

Heaping and rounding. — A potential concern is whether the distribution of haflaga values shows artificial heaping at round numbers (e.g., $H = 28$ or $H = 30$) due to rounding or reporting conventions in the NFP dataset. Note that the halachic haflaga convention $H_k = L_k + 1$ means the concern is about heaping in the raw cycle lengths L_k : rounding at $L = 28$ (haflaga $H = 29$) and $L = 30$ (haflaga $H = 31$). We examined the raw frequency distribution: $L = 27$ ($H = 28$, 194 cycles), $L = 28$ ($H = 29$, 225 cycles, the mode), $L = 29$ ($H = 30$, 175 cycles, the Week-Dilug anchor), $L = 30$ ($H = 31$, 141 cycles). The distribution peaks at $L = 28$, which reflects the known biological mode of the menstrual cycle, and declines monotonically through $L = 30$. There is no excess at $L = 28$ or $L = 30$ relative to neighbours, and no evidence of rounding or reporting artefacts. The haflaga anchor value $H = 30$ (corresponding to $L = 29$)

similarly shows no excess (175 cycles vs. 225 at $H = 29$ and 141 at $H = 31$). Figure 11 confirms the smooth distribution.

Day-night timing (onset period). — A central requirement of halachic vessel law is that a pattern may be established only if all its constituent menstruations arrived during the *same onset period* — either all during the daytime period or all during the nighttime period. The NFP dataset records cycle length in whole days but not the time of day of menstrual onset. This creates two types of misclassification: (i) patterns that are counted as established in our analysis may not qualify halachically because the onsets spanned both day and night; and (ii) patterns that are not counted may in fact qualify because the day-rounding convention used here differs from the halachic one. Both types of error introduce noise that attenuates the observed pattern frequencies. The net direction of misclassification bias is indeterminate without data on onset timing: sequences counted as established patterns may not qualify halachically (upward bias), while sequences that do qualify halachically may be missed (downward bias). The two effects partially cancel, but their relative magnitude cannot be assessed from the present data, and the limitation applies equally to all five pattern types. The misclassification bias is expected to be approximately uniform across women and across pattern types, since the NFP dataset provides no information about individual variation in onset-period timing.

Calendar dates. — The dataset does not record calendar dates, making it impossible to test the 10 date-dependent vessel types (e.g., *vessel hachodesh*, the monthly calendar anchor, and weekday-based patterns). Only the 7 cycle-length-based vessel types are accessible; of these, two (Dilug Chozer Chalila and Haflaga Chozer Chalila) had zero occurrences (mathematically: Dilug Chozer Chalila requires three consecutive Dilug sequences, necessitating a minimum run of ~ 9 cycles with a specific arithmetic structure; given that a single Dilug occurs in only $\sim 4\%$ of possible triples in this dataset, the expected count of three consecutive Dilug sequences is negligibly small; the same reasoning applies to Haflaga Chozer Chalila). The two patterns actually zero occurrences, leaving the 5 patterns analysed here.

Body-sign vessel. — The dataset does not include prodromal physical symptoms (*vessel haguf*, body-sign vessel: a category established when a woman experiences a consistent recurring

physical sensation — such as yawning, shivering, or lower-abdominal pain — immediately preceding menstrual onset). Women in the sample may have established body-sign vessel patterns that are entirely invisible to a cycle-length analysis.

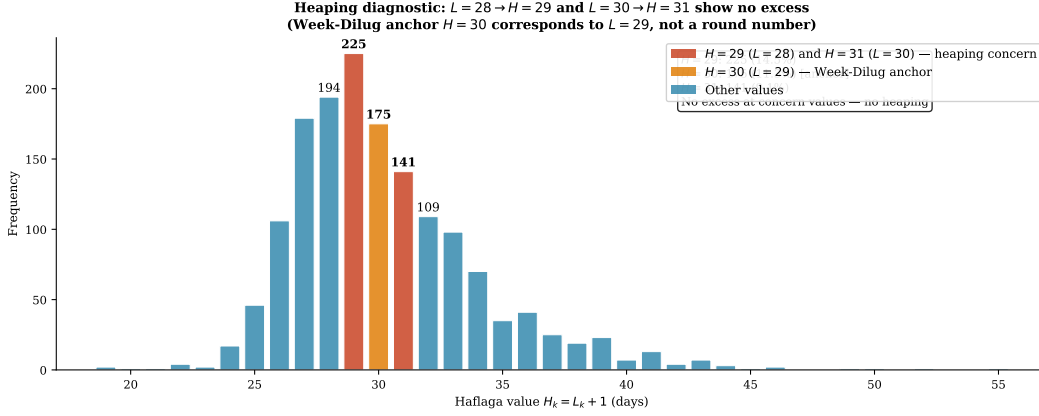


Figure 11: Distribution of haflaga values H_k around $H = 28$ and $H = 30$. The distribution is smooth with no heaping artefact at $H = 30$ ($H = 29$: 225 cycles; $H = 30$: 175; $H = 31$: 141), confirming that the Week-Dilug anchor at $H = 30$ does not reflect a reporting bias.

Population generalizability. — The participants are non-Jewish NFP users [6]: married women aged 18–42 using natural family planning to avoid or achieve pregnancy. The dataset excludes women in irregular hormonal states (menopausal, perimenopausal, postpartum); inclusion required regularly cycling women of reproductive age, so concerns about menopause or contraceptive washout are not applicable. Several halachic authorities hold that regularities observed in non-Jewish women may not generalise to Jewish women in practice [16]. More broadly, NFP users are a self-selected group who are likely more cycle-aware and may have more regular cycles than the general population, introducing a potential upward bias in regularity-dependent patterns such as Haflaga.

Null model structure. — The permutation and multinomial null models shuffle cycle lengths globally across women, ignoring within-person autocorrelation structure [1, 17]. The within-woman permutation null (H_W) addresses this by conditioning on each woman’s own cycle distribution and is therefore the more stringent test; future work could also employ parametric nulls based on fitted state-space models [1]. Additionally, the within-woman permutation null (H_W) tests the specific exchangeability null that all orderings of a woman’s observed cycle-length multiset are equally probable. This is narrower than a general test of serial independence,

stationarity, or absence of autocorrelation: a process with Markov-type autocorrelation or changing latent states could still fail to produce enough exact pattern excess to reject this particular randomisation null. The non-significant H_W result should therefore be stated as a failure to detect excess under within-woman exchangeability, not as the absence of all forms of temporal dependence. Future work could employ parametric nulls based on fitted state-space models [1, 17] to assess robustness to this limitation.

Pairwise dependence: cycle-count confound. — The per-woman binary association analysis (Table 3) is subject to a cycle-count confound: women with more cycles have more opportunities to establish any pattern, potentially inflating co-occurrence rates independently of the underlying process. Given the substantial variation in cycles per woman (range 1–45), associations involving rare patterns (Week-Dilug: 18 women; Dilug-in-Dilug: 23 women) should be interpreted with particular caution. The two significant pairs (Haflaga $\times$ Week-Dilug and Haflaga $\times$ Dilug-in-Dilug) are at least partly mathematically entailed (set-theoretic inclusion and polynomial proximity), which provides independent justification beyond the empirical association.

Sample size and power. — The requirement for minimum run lengths (3 cycles for Haflaga and Dilug; 4 for Dilug-in-Dilug) limits the number of testable windows, particularly for women with few recorded cycles. With a mean of 13.2 cycles per woman, longer patterns such as Dilug-in-Dilug are likely underpowered. As an approximate sensitivity analysis, we computed a minimum detectable count (MDC) for each pattern under H_W at nominal 80% power and $\alpha = .05$ (one-sided), using a normal-theory location-shift approximation:

$$\text{MDC} \approx \mu_W + (z_{0.05} + z_{0.80}) \sigma_W = \mu_W + 2.487 \sigma_W, \quad (2)$$

where μ_W and σ_W are the mean and standard deviation of the within-woman permutation null distribution (Table 4). This formula assumes that the alternative distribution is a location-shifted version of the null (same SD, different mean) and that both are approximately normal — assumptions that are not verified for this discrete pattern-count statistic. The MDC values should therefore be interpreted as approximate normal-theory benchmarks, not as exact power guarantees. A simulation-based power analysis under explicit temporal alternatives (e.g., a

Markov model with specified lag-1 transition strength) is deferred to future work with a larger dataset. All five observed counts fall below their respective MDC, consistent with the current sample ($N = 118$ women, 1,554 cycles) having limited power to detect individual pattern excess under H_W . Note that the joint Mahalanobis test under H_W is also non-significant ($D_M = 2.95$, $p = .121$), so underpowerment applies at the multivariate level as well. The significant Mahalanobis test ($D_M = 5.10$, $p < .001$) reported in Section 4.5 refers to the *global* permutation null (H_G), not the within-woman null (H_W); both results are genuine, but they answer different scientific questions: H_G tests overrepresentation relative to all women in the sample, while H_W tests whether temporal ordering within each woman’s own cycles adds overrepresentation beyond her individual cycle-length distribution.

Table 4: Approximate detectable-count benchmarks under H_W (within-woman permutation null), based on a normal-theory location-shift approximation. μ_W : null mean; σ_W : null SD; 95th pct: empirical 95th percentile of null (rejection threshold at $\alpha = .05$); MDC: minimum detectable count at nominal 80% sensitivity ($\text{MDC} \approx \mu_W + 2.487 \sigma_W$, where $z_{0.05} + z_{0.80} = 1.645 + 0.842 = 2.487$; approximate normal-theory benchmark, not an exact power guarantee); z_{obs} : standardised observed count. All observed counts fall below the MDC, consistent with limited sensitivity at $N = 118$.

Pattern	Obs.	μ_W	σ_W	95th pct	MDC	z_{obs}
Haflaga	26	31.85	4.68	40.0	44	−1.25
Dilug	61	67.07	7.36	79.0	86	−0.83
Week	32	33.81	3.88	40.0	44	−0.47
Week-Dilug	18	20.94	3.22	26.0	29	−0.91
Dilug-in-Dilug	27	39.14	5.86	49.0	54	−2.07

6. Conclusion

This study provides, to our knowledge, the first formal statistical test of whether halachic vessel patterns are overrepresented in empirical menstrual cycle data. The five patterns analysed form a polynomial hierarchy whose structure generates mathematically necessary dependencies, confirmed by exploratory pairwise randomisation-based dependence tests. Three complementary null models were applied: a global permutation null, a multinomial null, and the scientifically sharper within-woman permutation null (H_W).

The primary result is from the within-woman null: under H_W , *no pattern is individually significant* (all $p \geq .722$; joint $D_M = 2.95$, $p = .121$), and all observed counts fall below their within-woman null means. These findings provide no evidence, under within-woman

exchangeability, that temporal ordering within each woman’s cycle sequence creates vessel patterns beyond what her own cycle-length distribution predicts.

Under the global null, Haflaga is significantly overrepresented ($p < .001$ after Holm correction); Dilug is nominally overrepresented ($p = .043$) but does not survive Holm correction and should be treated as hypothesis-generating pending replication. The joint Mahalanobis test confirms significant displacement from the global null ($D_M = 5.10, p < .001$). Only 2 of 50,000 permutation simulations reached the observed Haflaga count. The primary focused joint test on the three structurally dependent patterns (Haflaga, Week-Dilug, Dilug-in-Dilug) — justified by set-theoretic containment and polynomial-hierarchy proximity prior to data inspection — yields $D_M = 4.52, p < .001$. The 5-pattern robustness check ($D_M = 5.10, p < .001$) confirms the result is not sensitive to pattern-subset choice. Together, these results suggest that halachic vessel rules function primarily as empirical detectors of individual cycle regularity: the rules identify the subset of women whose cycles are intrinsically stable and formalise that stability as an anticipatory pattern. The joint distance-based permutation test introduced here is a methodological contribution with potential applications beyond the halachic context. In particular, the Mahalanobis formulation offers a principled approach for composite hypothesis testing when multiple correlated count outcomes arise from shared biological structure: rather than applying multiple corrections to individually underpowered tests, it aggregates evidence into a single joint statistic that respects inter-pattern correlations. This may prove useful for similar hypothesis-testing problems in cyclic biological processes more broadly.

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